

Сетевые технологии

Лабораторная работа №3

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Информация

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Цель данной работы — изучение посредством Wireshark кадров Ethernet, анализ PDU протоколов транспортного и прикладного уровней стека TCP/IP.

```
PS C:\Users\nokt> ipconfig /all

Настройка протокола IP для Windows

Имя компьютера . . . . . : DESKTOP-2PD03M3
Основной DNS-суффикс . . . . . : dk.sci.pfu.edu.ru
Тип узла . . . . . : Гибридный
IP-маршрутизация включена . . . . . : Нет
WINS-прокси включен . . . . . : Нет

Адаптер Ethernet Ethernet:

Состояние среды. . . . . : Среда передачи недоступна.
DNS-суффикс подключения . . . . . : dk.sci.pfu.edu.ru
Описание . . . . . : Realtek PCIe GbE Family Controller
Физический адрес. . . . . : D4-93-90-10-S1-BF
DHCP включен. . . . . : Да
Автонастройка включена. . . . . : Да

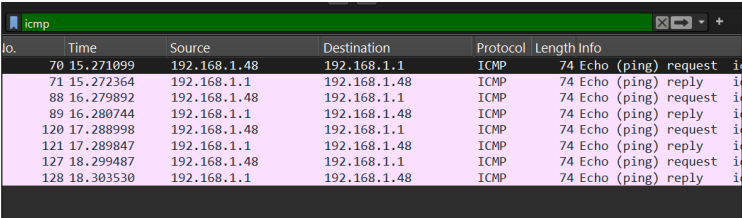
Адаптер беспроводной локальной сети Подключение по локальной сети* 2:

Состояние среды. . . . . : Среда передачи недоступна.
DNS-суффикс подключения . . . . . :
Описание . . . . . : Microsoft Wi-Fi Direct Virtual Adapte
г
Физический адрес. . . . . : DC-21-5C-2E-C6-70
DHCP включен. . . . . : Да
Автонастройка включена. . . . . : Да

Адаптер беспроводной локальной сети Подключение по локальной сети* 1:

Состояние среды. . . . . : Среда передачи недоступна.
DNS-суффикс подключения . . . . . :
Описание . . . . . : Microsoft Wi-Fi Direct Virtual Adapte
г #2
Физический адрес. . . . . : DE-21-5C-2E-C6-6F
DHCP включен. . . . . : Нет
Автонастройка включена. . . . . : Да
```

Рис. 1: ipconfig /all



The image shows a Wireshark packet capture window with the title bar 'icmp'. The packet list pane displays a series of ICMP Echo (ping) requests and replies. The first packet is a request from 192.168.1.48 to 192.168.1.1. Subsequent packets are replies from 192.168.1.1 to 192.168.1.48, followed by another request and then more replies. The packet details pane is empty.

No.	Time	Source	Destination	Protocol	Length	Info
70	15.271099	192.168.1.48	192.168.1.1	ICMP	74	Echo (ping) request i
71	15.272364	192.168.1.1	192.168.1.48	ICMP	74	Echo (ping) reply i
88	16.279892	192.168.1.48	192.168.1.1	ICMP	74	Echo (ping) request i
89	16.280744	192.168.1.1	192.168.1.48	ICMP	74	Echo (ping) reply i
120	17.288998	192.168.1.48	192.168.1.1	ICMP	74	Echo (ping) request i
121	17.289847	192.168.1.1	192.168.1.48	ICMP	74	Echo (ping) reply i
127	18.299487	192.168.1.48	192.168.1.1	ICMP	74	Echo (ping) request i
128	18.303530	192.168.1.1	192.168.1.48	ICMP	74	Echo (ping) reply i

Рис. 2: Wireshark after ping

```
* Frame 20: Packet: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface DeviceNPF_{20631ff6-B006-401C-A962-842A40D7ED09}
  Section number: 1
  * Interface id: 0 (Device\NPF_{20631ff6-B006-401C-A962-842A40D7ED09})
    Encapsulation type: Ethernet (1)
    Arrival Time: Feb 13, 2026 18:42:55.421756000 (omprys, russia)
    UTC Arrival Time: Feb 13, 2026 15:42:55.421756000 UTC
    Epoch Arrival Time: 170997775.421756000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 174.340000 milliseconds]
    [Time since reference or first frame: 15.271099000 seconds]
    Frame Number: 20
    Frame length: 74 bytes (592 bits)
    Capture length: 74 bytes (592 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:icmp:data]
    Character encoding: ASCII (0)
    [Coloring Rule Name: ICMP]
    [Coloring Rule String: icmp || icmpv6]
  * Ethernet II, Src: Intel_2c:c6:6f (dc:21:5c:2e:c6:6f), Dst: KemetiC_11:32:a7 (58:ff:20:11:32:a7)
    * Destination: KemetiC_11:32:a7 (58:ff:20:11:32:a7)
      ....:0b .... .... .... = 48 bit: Globally unique address (factory default)
      ....:0b .... .... .... = 48 bit: Individual address (unicast)
    * Source: Intel_2c:c6:6f (dc:21:5c:2e:c6:6f)
      Type: IPv6 (0x86dd)
      [Stream index: 0]
  * Internet Protocol Version 4, Src: 192.168.1.48, Dst: 192.168.1.1
```

Рис. 3: Ethernet2

Выполнение лабораторной работы

```
595 44.152486 188.184.67.127 192.168.1.48 HTTP 1104 HTTP/1.1 200 OK (text/html)
618 47.540117 188.185.48.40 192.168.1.48 HTTP 299 [TCP Retransmission] HTTP/1.1 408 Request Time-out (text/html)
620 47.542656 188.185.48.40 192.168.1.48 HTTP 299 [TCP Retransmission] HTTP/1.1 408 Request Time-out (text/html)

Frame 595: Packet, 1104 bytes on wire (8832 bits), 1104 bytes captured (8832 bits) on interface \Device\NPF_{70631EE6-8C80-461C-B962-842A4D07ED09}
Ethernet II, Src: Keenetic_11:32:a7 (58:ff:20:11:32:a7), Dst: Intel_2e:c6:6f (dc:21:5c:2e:c6:6f)
  Destination: Intel_2e:c6:6f (dc:21:5c:2e:c6:6f)
    ....0. .... .... = LG bit: Globally unique address (factory default)
    ....0. .... .... = 1G bit: Individual address (unicast)
  Source: Keenetic_11:32:a7 (58:ff:20:11:32:a7)
    ....0. .... .... = LG bit: Globally unique address (factory default)
    ....0. .... .... = 1G bit: Individual address (unicast)
  Type: IPv6 (80000)
  [Stream index: 0]
Internet Protocol Version 4, Src: 188.184.67.127, Dst: 192.168.1.48
Transmission Control Protocol, Src Port: 80, Dst Port: 55591, Seq: 4285, Ack: 560, Len: 1038
  Source Port: 80
  Destination Port: 55591
  [Stream index: 26]
  [Stream Packet Number: 10]
  [Conversation completeness: Complete, WITH_DATA (31)]
  [TCP Segment Len: 1038]
```

Рис. 4: Http

Выполнение лабораторной работы

191 33.488418	192.168.1.48	192.168.1.1	DNS	76 Standard query 0x8973 A home.web.cern.ch
192 35.723953	192.168.1.48	192.168.1.1	DNS	76 Standard query 0x8973 A home.web.cern.ch
194 35.823139	192.168.1.1	192.168.1.48	DNS	158 Standard query response 0x8973 A home.web.cern.ch CNAME drupal-apps-shard-1.co
195 33.851941	192.168.1.1	192.168.1.48	DNS	158 Standard query response 0x8973 A home.web.cern.ch CNAME drupal-apps-shard-1.co
196 33.851976	192.168.1.48	192.168.1.1	DNS	106 Destination unreachable (Port unreachable)
215 37.754865	192.168.1.48	192.168.1.1	DNS	93 Standard query 0xb0b6 A optimizationguide-pa.googleapis.com
216 37.761259	192.168.1.1	192.168.1.48	DNS	159 Standard query response 0xb0b6 A optimizationguide-pa.googleapis.com A 216.239
340 24.919579	192.168.1.48	192.168.1.1	DNS	80 Standard query 0xb0b6 A astrid.lickypn.net
341 24.931702	192.168.1.1	192.168.1.48	DNS	112 Standard query response 0xb0b6 A astrid.lickypn.net A 172.67.152.110 A 104.2
418 29.159791	192.168.1.48	192.168.1.1	DNS	76 Standard query 0x3483 A alive.github.com
419 29.171235	192.168.1.1	192.168.1.48	DNS	111 Standard query response 0x3483 A alive.github.com CNAME live.github.com A 140.
441 31.551779	192.168.1.48	192.168.1.1	DNS	78 Standard query 0xb0c47 A github.com
442 31.560783	192.168.1.1	192.168.1.48	DNS	86 Standard query response 0xb0c47 A github.com A 140.82.121.3
661 52.074382	192.168.1.48	192.168.1.1	DNS	83 Standard query 0xf8fc A rus2-1.web.telegram.org
662 52.080526	192.168.1.1	192.168.1.48	DNS	99 Standard query response 0xf8fc A rus2-1.web.telegram.org A 149.154.167.99
691 53.450361	192.168.1.48	192.168.1.1	DNS	74 Standard query 0xb310 A mail.yandex.ru
692 53.459363	192.168.1.1	192.168.1.48	DNS	68 Standard query response 0xb310 A mail.yandex.ru A 139.16.31.35

Frame 191: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface \Device\NPF_{7D631E16-8C8B-461C-A962-B42A007D095}, 14 0
Ethernet II, Src: Intel_Juic618f (dc:21:5c:2a:c6:0f), Dst: Keenetic_11:32:a7 (50:ff:20:11:32:a7)
Destination: Keenetic_11:32:a7 (50:ff:20:11:32:a7)
... .. 16 bit: Globally unique address (factory default)
... .. 16 bit: Individual address (unicast)
Source: Intel_Juic618f (dc:21:5c:2a:c6:0f)
... .. 16 bit: Globally unique address (factory default)
... .. 16 bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 192.168.1.48, Dst: 192.168.1.1
User Datagram Protocol, Src Port: 50236, Dst Port: 53
Source Port: 50236
Destination Port: 53
Length: 42
Checksum: 0xb3bd [unverified]
[Checksum status: Unverified]
[Stream index: 3]
[Stream Order Number: 11]

Рис. 5: dns

Выполнение лабораторной работы

No.	Time	Source	Destination	Protocol	Length	Info
191	13.684418	192.168.1.48	192.168.1.1	DNS	76	Standard query 0x9773 A home.web.cern.ch
192	13.722953	192.168.1.48	192.168.1.1	DNS	76	Standard query 0x9773 A home.web.cern.ch
194	13.825139	192.168.1.1	192.168.1.48	DNS	158	Standard query response 0x9773 A home.web.cern.ch CNAME drupal-apps-shard
195	13.851941	192.168.1.1	192.168.1.48	DNS	158	Standard query response 0x9773 A home.web.cern.ch CNAME drupal-apps-shard
196	13.901046	192.168.1.48	192.168.1.1	DNS	160	Standard query response 0x9773 A home.web.cern.ch CNAME drupal-apps-shard
215	17.757085	192.168.1.48	192.168.1.1	DNS	95	Standard query 0x800b A optimizationguide-pa.googleapis.com
216	17.763259	192.168.1.1	192.168.1.48	DNS	159	Standard query response 0x800b A optimizationguide-pa.googleapis.com A 236
340	24.919578	192.168.1.48	192.168.1.1	DNS	80	Standard query 0x0a6d A astrid.iclickops.net
341	24.933702	192.168.1.1	192.168.1.48	DNS	112	Standard query response 0x0a6d A astrid.iclickops.net A 172.67.152.110 A 1
418	29.159791	192.168.1.48	192.168.1.1	DNS	76	Standard query 0x3483 A alive.github.com
419	29.173235	192.168.1.1	192.168.1.48	DNS	111	Standard query response 0x3483 A alive.github.com CNAME live.github.com A
440	31.551779	192.168.1.48	192.168.1.1	DNS	70	Standard query 0xb047 A github.com
442	31.565783	192.168.1.1	192.168.1.48	DNS	86	Standard query response 0xb047 A github.com A 140.82.121.3
661	52.074302	192.168.1.48	192.168.1.1	DNS	83	Standard query 0xf8ce A 2uc2-1.web.telegram.org
662	52.080526	192.168.1.1	192.168.1.48	DNS	99	Standard query response 0xf8ce A 2uc2-1.web.telegram.org A 149.154.167.99
691	53.692561	192.168.1.48	192.168.1.1	DNS	74	Standard query 0xe310 A mail.yandex.ru
692	53.698309	192.168.1.1	192.168.1.48	DNS	89	Standard query response 0xe310 A mail.yandex.ru A 172.67.152.110
+ Frame 194: Packet, 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface \Device\NPF_{70631E16-BC80-461C-B962-842A40D7E009}, id 0						
+ Ethernet II, Src: Realtek_11:32:a7 (50:ff:20:11:32:a7), Dst: Intel_2e:c6:6f (dc:21:5c:2e:c6:6f)						
+ Destination: Intel_2e:c6:6f (dc:21:5c:2e:c6:6f)						
+ 00 0000 0000 = 16 bit: Globally unique address (factory default)						
+ 00 0000 0000 = 16 bit: Individual address (unicast)						
+ Source: Realtek_11:32:a7 (50:ff:20:11:32:a7)						
+ 00 0000 0000 = 16 bit: Globally unique address (factory default)						
+ 00 0000 0000 = 16 bit: Individual address (unicast)						
Type: IPv4 (0x0800)						
[Stream Index: 4]						
+ Internet Protocol Version 4, Src: 192.168.1.1, Dst: 192.168.1.48						
+ User Datagram Protocol, Src Port: 53, Dst Port: 50236						
+ Source Port: 53						
+ Destination Port: 50236						
+ Length: 124						
+ Checksum: 0xf7ed [unverified]						
+ [Checksum Status: Unverified]						
+ [Stream Index: 3]						
+ [Stream Packet Number: 33]						

Рис. 6: dns-ответ

Выполнение лабораторной работы

```
1 3 0.251063 143.154.167.93 552 556 1.48 TCP 66.443 → 66.400 [ACK] Seq=1431108 Win=1638 len=0 (Win=2271) Window=200
1 4 0.262067 143.154.167.93 552 556 1.48 15.561.2 278 Application Data

+ Transmission Control Protocol, Src Port: 443, Dst Port: 46400, Seq: 1, Ack: 104, Len: 0
  Source Port: 443
  Destination Port: 46400
  [Stream index: 0]
  [Stream Packet Number: 2]
  [TCP Segment len: 0]
  [Conversation completeness: Incomplete (12)]
  [IP Segment len: 0]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 29437479
  [Next Sequence Number: 1 (relative sequence number)]
  Acknowledgment Number: 104 (relative ack number)
  Acknowledgment Number (raw): 26172333
  3000 .... = Header Length: 32 bytes (0)
  [Flags: Reset (RST)]
    RST .... = Reset: Not set
    ...F .... = Fin: Not set
    ...O .... = Opaque RST: Not set
    ....C .... = Connection Window Reduced: Not set
    ....R .... = RST (set): Not set
    ....U .... = Urgent: Not set
    ....I .... = Acknowledgment: Set
    ....A .... = Push: Not set
    ....R .... = Reset: Not set
    ....S .... = Syn: Not set
    ....B .... = Fin: Not set
  [IP Flags: ----->-----]
  Window: 3436
  [Calculated window size: 1638]
  [Window size scaling factor: 1 (unknown)]
  Checksum: 0x1a06 [correct]
  [Checksum status: correct]
  Urgent Pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
```

Рис. 7: tcp handshake

Выполнение лабораторной работы

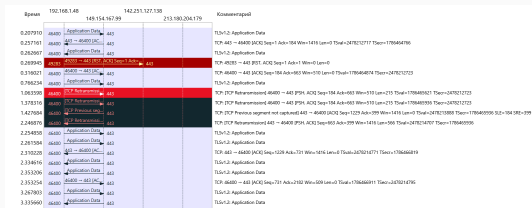


Рис. 8: График потока

В ходе данной лабораторной работы мной были изучены посредством Wireshark кадры Ethernet, получены практические навыки по анализу PDU протоколов транспортного и прикладного уровней стека TCP/IP.